Andrew Shao

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Education

University of Michigan Ann Arbor, MI
Bachelor of Science in Engineering: Computer Science Aug 2022 - Dec 2025

- **Relevant Coursework:**
  Distributed Systems, Web Systems, Operating Systems, Applied Parallel Programming with GPUs, Machine Learning, Natural Language Processing, Applied Linear Algebra

Work Experience

Rockwell Automation | *Software Intern* May 2025 - Aug 2025

- Designed and deployed an asynchronous event-driven pipeline using Apache Kafka and Python to process real-time telemetry data for debug purposes from virtualized controllers, increasing data ingestion throughput by 35%.
- Navigated a large legacy codebase for Logix Designer to resolve 10+ critical concurrency bugs in C++ and C; utilized .NET profiling through Visual Studio optimizing asynchronous resource management, reducing interface latency.
- Built an automated security tool in Python using GitLab CI/CD that integrates with the BlackDuck API to parse security vulnerability data and generate Jira tickets, streamlining the reporting workflow and improving triage efficiency by 20%.

Sigwise | *Software Intern* May 2023 - Jul 2023

- Developed a video storage Android application using Java, implementing an LRU caching mechanism and integrating with the Google Cloud Storage (GCS) API to ensure low-latency video playback, reducing load times by 20%.
- Designed and deployed Docker containers for backend alert services, utilizing Kubernetes for cluster management, scaling, and creating consistent development environments to streamline testing.

Hyland Software | *Software Intern (PEEKE Internship Program)* Jun 2021 - Sept 2021

- Created a full-stack web application to host the next Hyland Hackathon using React.js, Django, and SQLite, designed to display real-time announcements, chat messages, and team data, allowing the hackathon to shift to a hybrid format.
- Developed and executed unit and integration test cases using Pytest and Jest to ensure code quality and reliability.

University of Michigan | *Computer Consultant* Oct 2022 - Mar 2026

- Provided classroom technology assistance for computers, projectors, and Zoom software to the UofM SEAS building.
- Built a Chrome browser extension in Javascript that restructured helpdesk ticket data into a condensed view and automated repetitive workflow actions via custom macros, reducing time spent triaging tickets by 15%.

Projects

Distributed Job Scheduler *Go, gRPC, etcd/Raft*

- Developed a distributed job scheduler in Go using gRPC for inter-node communication and bin-packing to allocate jobs across simulated worker nodes by CPU/memory, with priority classes and preemption for resource contention.
- Implemented heartbeat-based health checks for fault-tolerant failure detection and job rescheduling, backed by Raft consensus (etcd) for scheduler state, as well as per-namespace quota enforcement and autoscaling.

Distributed Key-Value Store (Paxos & Sharded KV Store) *C++, C*

- Built a fault-tolerant, horizontally scalable distributed key-value store in C++ using Paxos consensus and a sharded architecture, ensuring linearizable consistency and live shard migration.
- Designed a resilient RPC communication layer with idempotent request handling and optimized routing to guarantee at-most-once semantics and minimize end-to-end latency.

MapReduce Framework & Search Engine *Python, Flask, ReactJS, Javascript, Java, HTML, CSS*

- Developed a distributed MapReduce framework in Python using socket programming, multithreading, and a Manager-Worker architecture to enable fault-tolerant parallel data processing with worker monitoring and dynamic load balancing.
- Built a multi-stage data mining pipeline that processed thousands of crawled HTML documents into an inverted index, computing tf-idf, document normalization, and PageRank metrics for scalable search engine information retrieval.

Skills

Languages: C++, Go, C, Python, Java, C#, SQL, Bash, JavaScript
Technologies & Cloud: Docker, Kubernetes, GCP, AWS, CUDA, Linux/Unix, CI/CD (Jenkins), Kafka, Flask, React